Intervention Selection & Risk Classification

Intervention selection is guided by causal findings and system risk classification, aligned with regulatory requirements for high-risk AI systems.

Purpose

This document defines how to select the appropriate fairness intervention strategy based on:

- The causal source of bias
- The system's risk classification
- Operational constraints

The goal is to ensure intervention choices are evidence-based, proportionate, and governed — not ad hoc.

Core Principle

Fairness interventions must be selected based on where bias enters or is amplified, as identified in the Causal Fairness Analysis.

The intervention stage must correspond to the stage where bias:

- Originates
- Persists
- Or causes measurable harm

Selecting an intervention without a causal justification risks:

- Masking root causes
- Creating superficial compliance
- Introducing new trade-offs without fairness gains

Intervention selection follows a root-cause-first hierarchy: upstream corrections (data and model-level fixes) are preferred over downstream mitigation whenever technically feasible.

Primary Decision Factors

Intervention choice depends on four structured factors:

A. Causal Source of Bias

The Causal Fairness Analysis must identify whether bias enters at:

- Data Stage (representation imbalance, measurement error, biased labels)
- Model Stage (proxy learning, unequal error rates, amplification effects)
- Decision Stage (threshold effects, reject-option bias, deployment logic)

The intervention must correspond to the stage where bias enters or is amplified.

B. System Risk Classification

High-Risk Systems

- All intervention stages must be evaluated.
- Skipping stages requires formal governance approval.
- Validation and documentation are mandatory before deployment.
- Layered intervention is required if bias exists at multiple stages.

Lower-Risk Systems

- Targeted intervention is permissible if bias is isolated.
- Stages may be skipped only with documented justification.
- Validation remains required but may be proportionate.

C. Technical and Operational Constraints

Intervention choice may be influenced by:

- Whether the model can be retrained
- Whether the model is third-party or locked
- Data availability
- Deployment constraints

Constraints may justify temporary mitigation, but do not eliminate the need for causal documentation.

D. Governance & Compliance Requirement

Intervention selection must comply with governance and regulatory requirements.

For high-risk systems:

- All intervention stages must be evaluated
- Trade-offs must be explicitly documented
- Approval must be obtained before deployment

Failure to meet governance requirements invalidates the intervention decision.

When to Select Each Intervention

Select Pre-Processing When:

- Bias originates in the training data.
- Certain groups are underrepresented.
- Measurement quality differs systematically across groups.
- Labels reflect historical or structural bias.
- The causal graph shows bias entering before model training.

Pre-processing addresses what the model learns from.

Select In-Processing When:

- Disparities persist after data correction.
- The model learns proxy relationships.
- Error rates differ significantly across groups.
- Fairness constraints must be enforced during optimization.

In-processing addresses how the model learns.

Select Post-Processing When:

- Harm occurs near the decision threshold.
- Immediate mitigation is required.
- Retraining is infeasible or slow.
- The model cannot be modified directly.

Post-processing addresses how decisions are applied.

Intervention selection follows a structured hierarchy. Pre-processing is preferred when bias originates in data because it corrects structural causes before model learning occurs. In-processing is appropriate when bias arises from optimization behavior or persists after data correction. Post-processing is reserved for decision-stage harm or urgent mitigation scenarios and should not substitute upstream correction when retraining is feasible. Intervention selection must always be traceable to the causal analysis output to ensure alignment between identified bias and applied mitigation.

Layered Intervention Rule

If causal analysis identifies bias at multiple stages:

- Apply pre-processing to correct data imbalance or measurement bias.
- Apply in-processing to control model amplification.
- Apply post-processing to mitigate immediate decision harm.

Single-stage intervention is insufficient for multi-stage causal pathways.

Risk Classification Framework

Before selecting an intervention strategy, the system must be classified.

High-Risk Systems

Examples:

- Immigration decisions
- Credit approval
- Hiring or promotion systems
- Healthcare eligibility

Characteristics:

- Significant impact on rights or opportunities
- Regulatory scrutiny
- Potential for irreversible harm

High-risk systems require stricter procedural guarantees.

Lower-Risk Systems

Examples:

- Content personalization
- Non-critical recommendations
- Internal workflow optimization

Characteristics:

- Limited rights impact
- Reversible outcomes
- Lower regulatory exposure

Lower-risk systems allow proportional flexibility.

Intervention Decision Tree

Use this decision tree after completing Causal Fairness Analysis.

START
↓
Conduct Causal Fairness Analysis
↓
Where does bias primarily enter the system?

A. Bias enters at the Training Data / Measurement Stage

Examples:

- Data imbalance
- Underrepresentation
- Label bias
- OCR / NLP performance differences

→ Select Pre-Processing Intervention

Actions:

- Rebalance dataset
- Correct measurement error
- Adjust sampling or weighting
- Remove biased labels

If the system is High-Risk:

- The model and decision stages must still be evaluated.
- Validation is mandatory.

B. Bias enters at the Model Computation Stage

Examples:

- Proxy feature learning
- Unequal error rates
- Score calibration disparities
- Bias persists after data correction

→ Select In-Processing Intervention

Actions:

- Apply fairness constraints
- Modify loss function
- Retrain with fairness-aware optimization
- Reduce proxy amplification

If the system is High-Risk:

- Pre- and post-stages must still be evaluated.
- Trade-offs must be documented.

C. Bias enters at the Decision Threshold Stage

Examples:

- Threshold amplifies disparities
- Reject-option bias
- Harm is concentrated near the cutoff
- The model cannot be retrained

→ Select Post-Processing Intervention

Actions:

- Adjust decision thresholds
- Introduce review buffer zones
- Apply reject-option fairness
- Modify deployment logic

If harm is urgent:

- Post-processing may be applied immediately.
- Root-cause intervention must follow.

D. Bias enters at Multiple Stages

If causal analysis shows:

Data bias
AND
Model amplification
AND/OR
Threshold amplification

→ Apply Layered Intervention

- Pre-processing for data bias

- In-processing for model bias
- Post-processing for decision harm

Layered intervention is mandatory when causal pathways are multi-stage.

Mermaid Diagram:

A[Bias Detected] --> B{Where does bias occur?}

B -->|Data / Measurement| C[Pre-Processing]

B -->|Model Behavior| D[In-Processing]

B -->|Decision Threshold| E[Post-Processing]

B -->|Multiple Stages| F[Layered Intervention]

C --> G[Apply Data Fixes]

D --> H[Apply Model Constraints]

E --> I[Adjust Thresholds / Human Review]

F --> J[Combine All Interventions]

Decision Accountability

Every intervention decision must include:

- Identified bias stage
- Selected intervention type
- Justification for stage selection
- Explanation for skipped stages
- Risk classification reference

This ensures traceability and audit readiness.

How to Use This Decision Logic

1. Conduct causal analysis.
2. Identify where bias enters.
3. Determine system risk classification.
4. Follow the appropriate intervention pathway.

High-Risk Systems (e.g., IVRAS)

- Causal analysis is mandatory.
- All three intervention stages must be evaluated.
- Skipping requires governance approval.
- Validation and documentation cannot be bypassed.
- Layered intervention is required if bias exists at multiple stages.

Lower-Risk Systems

- Causal analysis recommended but may be scoped.
- If no bias is detected at a stage, that intervention may be skipped.
- Skipping requires a documented rationale.
- Post-processing may be used for rapid mitigation if proportional.

When Steps Can Be Skipped

An intervention stage may be skipped only if:

- Causal analysis shows no bias at that stage.
- Metrics confirm parity.
- Risk level allows proportional flexibility.
- Rationale is formally documented.

When Steps Cannot Be Skipped

A stage cannot be skipped when:

- The system is classified as high-risk.
- Harm has been observed.
- There is uncertainty about causal pathways.
- Regulatory or governance requirements mandate evaluation.

If a Stage Cannot Be Skipped

The team must:

1. Document why intervention is required.
2. Select the appropriate method.
3. Evaluate trade-offs.
4. Validate results.
5. Obtain formal approval before deployment.

Urgency Rule

If harm is immediate or severe:

- Post-processing may be applied first to reduce the impact
- Root-cause fixes (pre- or in-processing) must follow
- Temporary mitigation must not replace structural correction

Urgent interventions must be treated as temporary mitigation and formally tracked until permanent resolution is implemented.

Governance Requirements

No intervention decision is valid unless it is:

- Causally justified
- Proportionate to system risk
- Validated with defined fairness metrics
- Fully documented before deployment

If it is not documented, it is not approved.

Connection to Validation Framework

All selected interventions must proceed to the Validation and Monitoring Framework.

Validation ensures:

- Fairness improvement is measurable
- Performance trade-offs are acceptable
- No new bias is introduced

No intervention is considered complete until validation is successfully passed.